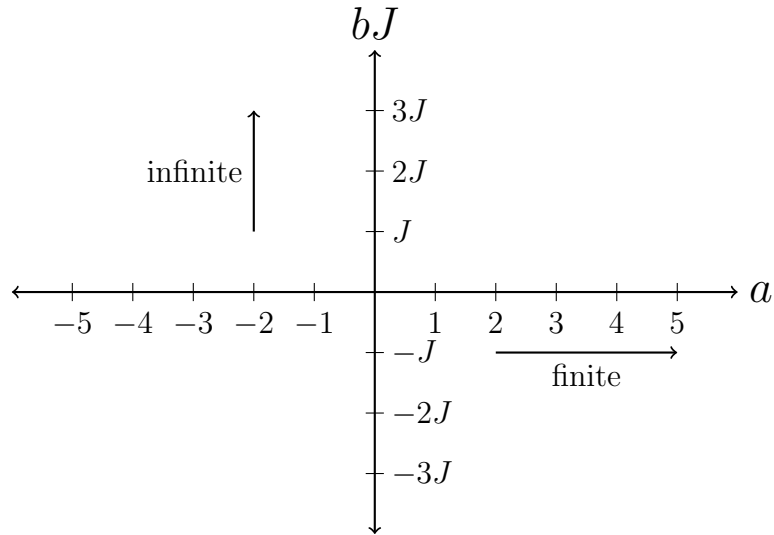


THE J-MATH

Let $\boxed{\frac{1}{0} = J}$

Let there be a 2D space, such that:



It is expressed in the form of

$$“a + bJ” \rightarrow \text{J-numbers}$$

$$\{a, b \in \mathbb{R}\}$$

$$a = \text{finite part}$$

$$bJ = \text{infinite part}$$

$$0 \cdot (a + bJ) = b$$

Addition

Addition is performed **component-wise**. Let there be two J-numbers, x and y , such that:

$$x = a + bJ$$

$$y = c + dJ$$

Then, the sum is:

$$x + y = (a + bJ) + (c + dJ)$$

$$x + y = (a + c) + (b + d)J$$

Exponential/Power Rules

Since $J = \frac{1}{0}$, we establish the following rules:

1. For $n > 0$:

$$J^n = \left(\frac{1}{0}\right)^n = J = \frac{1}{0}$$

2. For $n = 0$:

$$J^0 = \left(\frac{1}{0}\right)^0 = 1$$

3. For $n < 0$:

$$J^n = \left(\frac{1}{0}\right)^n = 0$$

Multiplication [Piecewise]

Let there be two J-numbers, x and y , such that:

$$x = a + bJ$$

$$y = c + dJ$$

Then, the product $x \cdot y$ is defined as:

$$x \cdot y = \begin{cases} b & \text{if } y = 0 \quad [i.e., c = 0 \text{ and } d = 0] \\ d & \text{if } x = 0 \quad [i.e., a = 0 \text{ and } b = 0] \\ ac + (ad + bc + bd)J & \text{otherwise} \end{cases}$$

Standard Real Numbers as J-Numbers

All standard real numbers can be written as J-numbers, such that their J-coefficient (b) is zero.

$$\begin{aligned} 0 &= 0 + 0J \\ 7 &= 7 + 0J \\ -5 &= -5 + 0J \end{aligned}$$

Important Note

One thing to be noted is that:

$$0 \cdot J \neq 0J$$

NOTE

1. **Commutativity** is intact.
2. **Distributive property** is abandoned conditionally. Its conditions are as follows:

$$X \cdot (Y + Z) \neq X \cdot Y + X \cdot Z$$

This applies if and only if all these following conditions are met:

- The multiplier (X) has a non-zero J-coefficient (b) .
- The inside terms $(Y + Z)$ sum to exactly absolute zero.

Otherwise, the distributive property holds and is intact.

3. **Associative property** is abandoned conditionally. Its conditions are as follows:

$$(X \cdot Y) \cdot Z \neq X \cdot (Y \cdot Z)$$

If and only if all these following conditions are met:

- An Absolute zero is involved.
- A J-number is involved.
- That J-number has a non-zero coefficient (b) .

Otherwise, the Associative property holds and is intact.

The Universal Golden Formula

Proof (1): Linear Derivation

Let there be a function of x , such that it is in the form of $a + bJ$ for some variable x :

$$F(x) = A + Bx$$

where $\{A, B, x \in \text{J-numbers}\}$

Let:

$$A = a_1 + a_2J$$

$$B = b_1 + b_2J$$

Let x be a variable in the form of $a + bJ$.

Evaluate $F(J)$

$\Rightarrow$ substitute $x \rightarrow 0 + 1J$

$$F(0 + 1J) = [a_1 + a_2J] + [(b_1 + b_2J) \cdot (0 + 1J)]$$

Using rules of multiplication established earlier:

$$F(0 + 1J) = [a_1 + a_2J] + [b_1(0) + (b_1(1) + b_2(0) + b_2(1))J]$$

$$F(0 + 1J) = [a_1 + a_2J] + [0 + (b_1 + 0 + b_2)J]$$

Using rules of addition established earlier:

$$F(0 + 1J) = (a_1 + a_2J) + (b_1 + b_2)J$$

$$\boxed{F(0 + 1J) = a_1 + (a_2 + b_1 + b_2)J} \quad (1)$$

OR

$$\boxed{F(J) = F(1) + [F(1) - F(0)]J} \quad (2)$$

(for $b_2 = 0$)

$F(x) = A + Bx$ is a linear J-function.

Example: $F(x) = x^2$

$$F(0) = 0^2 = 0$$

$$F(1) = 1^2 = 1$$

$$\begin{aligned} F(J) &= F(0) + [F(1) - F(0)]J \\ &= 0 + [1 - 0]J \\ &= J \end{aligned}$$

Proof (2): Infinite Series Derivation

Let there be a function of x , such that:

$$F(x) = \sum_{n=0}^{\infty} C_n x^n$$

$$\text{At } F(0) = C_0 \tag{1}$$

$$\text{At } F(1) = C_0 + C_1 + C_2 + C_3 + \dots$$

$$\text{Now, } F(1) - F(0) = C_1 + C_2 + C_3 + \dots \tag{2}$$

Evaluation at $x = J$

Put $x = J$ in $F(x) = \sum_{n=0}^{\infty} C_n x^n$:

$$F(J) = C_0 + C_1 J^1 + C_2 J^2 + C_3 J^3 + C_4 J^4 + \dots$$

Since $J^n = J$ for $n > 0$:

$$F(J) = C_0 + C_1 J + C_2 J + C_3 J + C_4 J + \dots$$

$$F(J) = C_0 + (C_1 + C_2 + C_3 + C_4 + \dots)J$$

From Eq (1) and Eq (2):

$$\boxed{F(J) = F(0) + [F(1) - F(0)]J} \tag{3}$$

Proof (3): Taylor Series Derivation

Let there be a function of x such that:

$$F(x) = A + Bx$$

$\Rightarrow$ A Linear J-function.

Using the Taylor Series:

$$F(x) = \sum_{n=0}^{\infty} \frac{F^{(n)}(a)}{n!} (x - a)^n$$

For our work, we center the series at $a = 0$:

$$F(x) = \sum_{n=0}^{\infty} \frac{F^{(n)}(0)}{n!} x^n$$

Using the Taylor series for our Linear J-function:

$$F(x) = A + Bx$$

Because this function is linear, the 1st derivative is constant ($F'(x) = B$) and all higher-order derivatives are zero (since $J^n = J$ for $n > 0$).

Now, this becomes:

$$\boxed{F(x) = F(0) + F'(0)x} \quad \text{--- (1)} \tag{4}$$

$$F'(0) = \frac{F(x_2) - F(x_1)}{x_2 - x_1}$$

In our standard anchors $x_2 = 1$ and $x_1 = 0$, we get:

$$F'(0) = \frac{F(1) - F(0)}{1 - 0}$$

$$F'(0) = F(1) - F(0) \quad \text{--- (2)} \tag{5}$$

Put Eq. (2) in Eq. (1):

$$F(x) = F(0) + [F(1) - F(0)]x$$

Since at $x = J$, it becomes $x = 0 + 1J$:

$$F(0 + 1J) = F(0) + [F(1) - F(0)](0 + 1J)$$

OR

$$\boxed{F(J) = F(0) + [F(1) - F(0)]J} \tag{6}$$

Sequence and Series Analysis

Sequence Analysis

Let there be a sequence a_n , such that it is treated as a function of n :

$$f(n) = a_n$$

Then, using the UGF, the infinite-th term is:

$$\begin{aligned} f(J) &= f(0) + [f(1) - f(0)]J \\ \boxed{f(J) &= a_0 + [a_1 - a_0]J} \end{aligned} \tag{7}$$

Series Analysis

Let there be a series, such that its sum up to the term n is:

$$S_n = \sum_{k=0}^n a_k$$

Then, using the UGF:

$$\begin{aligned} S_J &= S(0) + [S(1) - S(0)]J \\ S_J &= a_0 + [(a_0 + a_1) - a_0]J \\ \boxed{S_J &= a_0 + a_1J} \end{aligned} \tag{8}$$

Absolute Zero

$$\text{Absolute Zero} = 0 + 0J \quad [\text{i.e., the origin}]$$

Since $J = \frac{1}{0}$, J represents infinity.

Since J is the inverse of zero, then zero is the inverse of J :

$$\text{If } J = \frac{1}{0} = 0^{-1}$$

$$\text{Then } 0 = \frac{1}{J} = J^{-1}$$

Origins of Infinity

Since $5J$ is different from $2J$ (i.e., $5(\text{infinity})$ is different from $2(\text{infinity})$):

$$\text{Then } 5J \neq 2J$$

An infinity originated from 5 is different from an infinity originated from 2. J 's (or infinities) remember their origin.

Types of Infinities and Zeros

There are different types of infinities (J 's). The same is valid for zero (J^{-1}):

$$5J^{-1} \neq 2J^{-1}$$

i.e., $5(\text{zero})$ is different from $2(\text{zero})$.

Then J -inverses (or zeros) remember their origin:

$$5J^{-1} \neq 2J^{-1} \quad [\text{Information wise}]$$

A zero originated from 5 is different from a zero originated from 2.

$\therefore$ There are different types of Zeros (J -inverses)

Absolute Infinity

$$\text{Absolute Infinity} = 0 + 1J$$

NOTE: While these equations evaluate to the same Absolute Zero, or the same Absolute Infinity, the lineage, or history of how they got there, is different. *Information is preserved.*

The Universal Golden Formula (for J)

Let there be a function of x , such that it is a linear J-function:

$$F(x) = A + Bx$$

where $\{A, B, x \in \text{J-numbers}\}$

Evaluate $F(a + bJ)$

Substitute $x \rightarrow a + bJ$:

$$F(a + bJ) = A + B(a + bJ)$$

As found earlier:

$$F(x_{extreme}) = F(0) + [F(1) - F(0)]x_{extreme}$$

Put $x_{extreme} = a + bJ$:

$$F(a + bJ) = F(0) + [F(1) - F(0)](a + bJ)$$

$$F(a + bJ) = F(0) + a[F(1) - F(0)] + b[F(1) - F(0)]J \quad \text{--- (1)}$$

Since:

$$F(a) = F(0) + a[F(1) - F(0)] \quad \text{--- (2)}$$

Put Eq. (2) in Eq. (1):

Completion of Universal Golden Formula

Substituting Eq. (2) into Eq. (1):

$$F(a + bJ) = F(a) + b[F(1) - F(0)]J$$

The Universal Golden Formula

$$\boxed{F(a + bJ) = F(a) + b \cdot \text{change} \cdot J} \quad (9)$$

Where:

- $F(a)$ is the evaluation of the finite part.
- b is the scaling factor of the J-coefficient.
- $\text{change} = [F(1) - F(0)]$ represents the fundamental gradient/delta of the function.

This formula allows us to evaluate any function F at an infinite coordinate $(a + bJ)$ by only knowing the behavior of the function at the finite points 0 and 1.

End of Framework Definition
